



Customer-Driven Energy Solutions

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FOUR SEASONS

HOTEL

DOHA

EXECUTIVE SUMMARY ENERGY AUDIT REPORT

January 2022

1. Introduction

Following an agreement with the management, Energy Efficiency Group **EEG** has undertaken an Ashrae Level II energy audit of the Four Seasons Hotel Doha, Qatar (abbreviated by **FSD**).

The Four Seasons Hotel Doha is a luxury hotel that first started operation in April 2005. It consists of a complex that includes the hotel's main building, a SPA building, a town House as well as two residential towers which are not in the scope of this report. The total built-up area of the hotel is 48,288m² and it comprises a total of 232 rooms .

A complete renovation was undertaken between 2018 and 2020 with full deep rooms renovations as well as other in other areas in the property, although no major MEP upgrades occurred.

A central utility plant which is located in the basement of Ooredoo tower built in the vicinity of the complex provides chilled water to all the complex. The technical floor including the chillers plant room and all other technical rooms has a built-up area of 1,540 m² excluding the Cooling Towers' external location.

2. Utilities Costs

FSD's utility sources are Electricity, LPG, and water.

The twelve-month baseline period used for the energy audit spans from [October 2020 to September 2021](#).

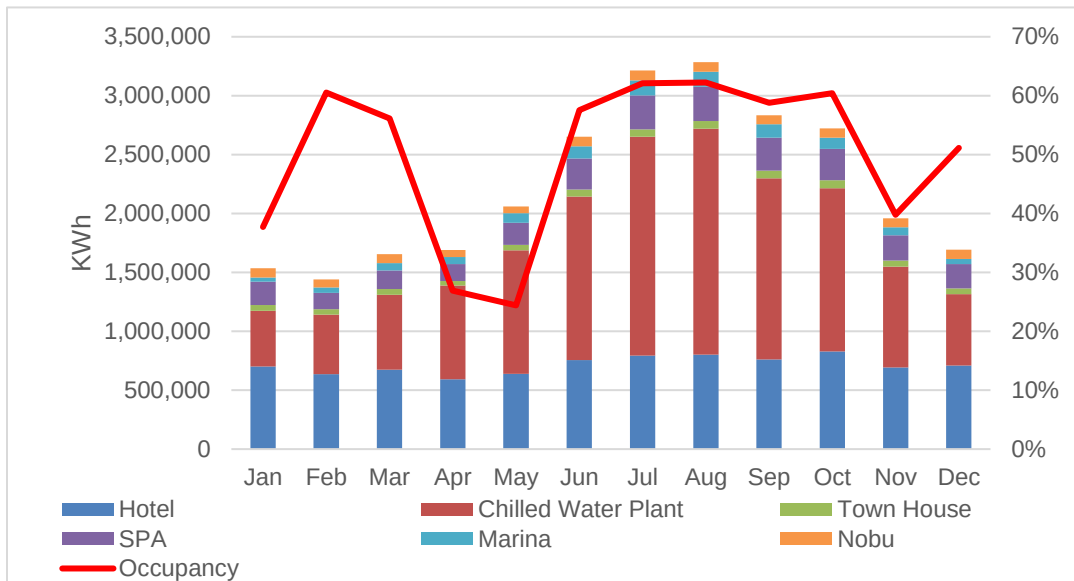
Electricity

Electricity is FSD is supplied by the local electricity supplier Kahramaa and serves the whole complex through an incoming feeder of 11 KV that supplies 16 step-down transformers covering all the buildings. There are 2 transformers for each of the residential towers, and since they are not in the scope of this report, it means that FSD has a total of 14 transformers. In addition, there are two 1250 KVA generators used in case of power failure and serving all the buildings. FSD is charged 75% of the Central Utility Plant's total electricity consumption with the remaining covered by the residential towers.

Electricity charges are at QAR 4,010,615 representing **75%** of the total energy & water-related utilities expenditures.

The chart in the following page provides the monthly electricity consumption of all the main components of FSD with the occupancy rate.

Figure 1: FSB Electricity Monthly Variations

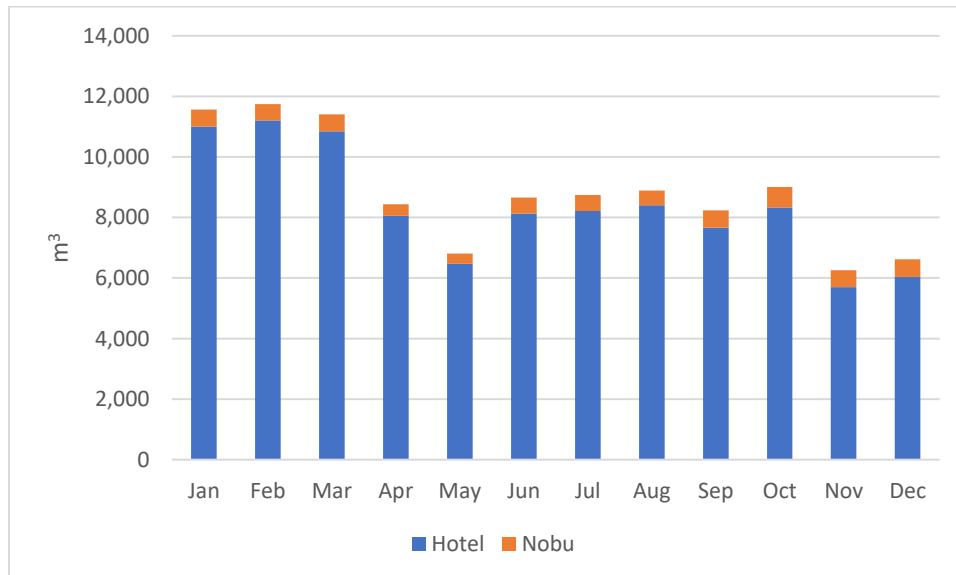


LPG

LPG is purchased in FSD and used for the kitchen equipment, as well as driving the six hot water boilers located in the 2nd floor's boiler room, and the two steam boilers located in the hotel's lower level. The LPG is also used in the Marina for supplying the boats.

Gas charges are at QAR 208,503 representing **4%** of the total utilities expenditures.

Figure 2: LPG Monthly Variations



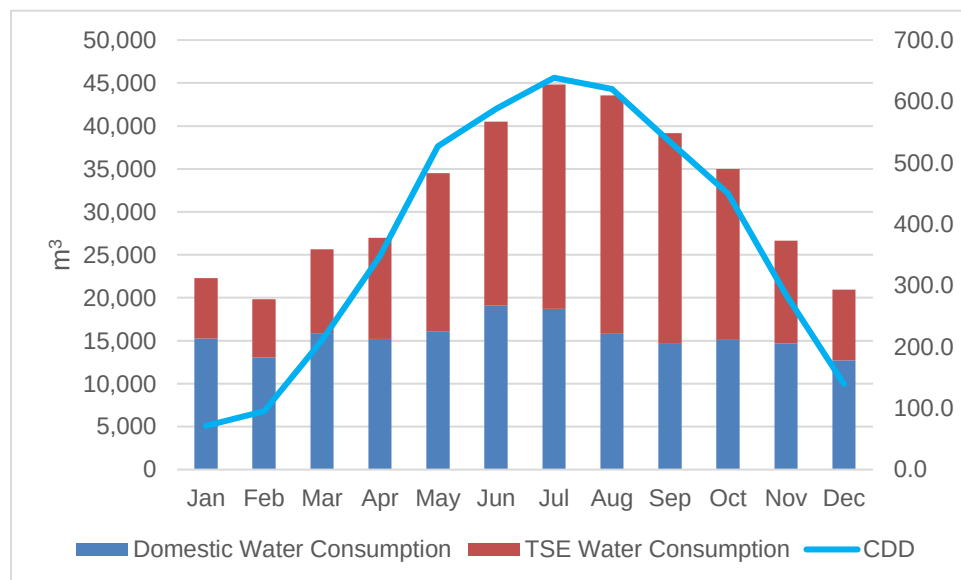
Water

Water in FSD is supplied by Kahramaa through connections to the water service infrastructure into the water storage tanks supplying the hotel, the ballroom building, the SPA, and the town house. Afterwards, it undergoes treatment before being delivered to the facility.

In addition, one of the key water end-uses in the complex are the cooling towers located in the central utility area. However, water is supplied to those towers through treated wastewater (TSE) from Kahramaa into two water storage tanks. This water goes through a Reverse Osmosis (RO) additional treatment before being used and is also used for irrigation.

Water charges are at QAR 1,129,263 per year representing **21%** of the total utilities cost.

Figure 3: Water Consumption Variations



Total Utilities

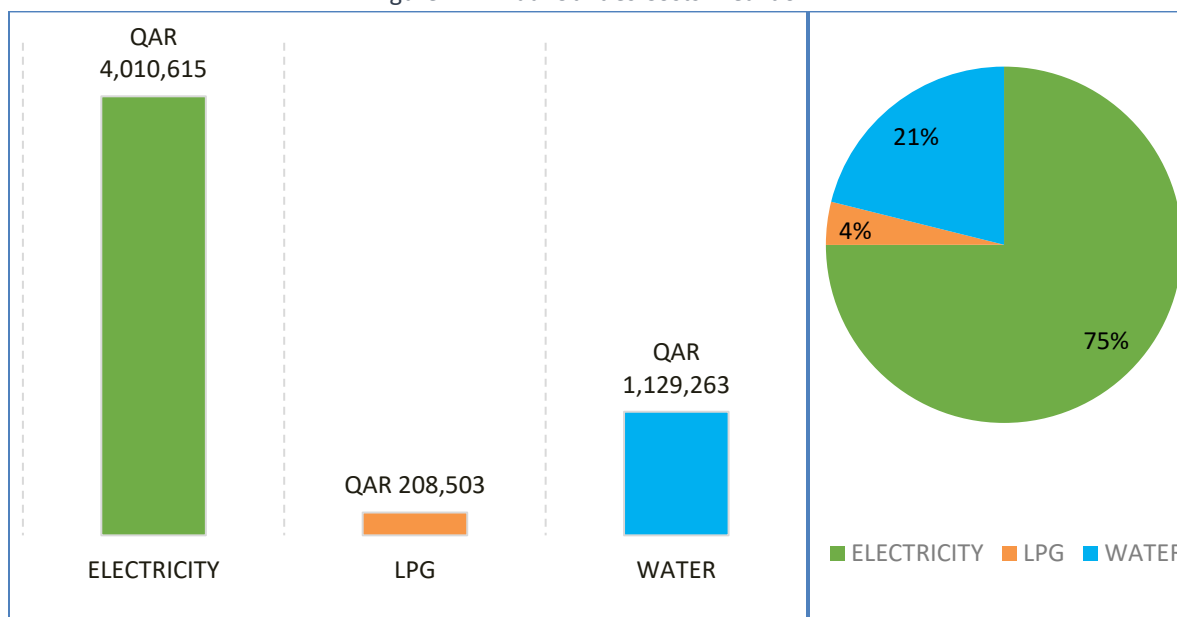
The below table provides the cost of each utility supply as compiled for the baseline year.

Total utilities cost is the aggregate of the cost of all utilities over the baseline period and they are shown in the following table and chart. As can be seen, electricity accounts for 75%, water for 21%, and LPG for 4%.

Table 1: Total Utilities Accounting Breakdown

UTILITY	ELECTRICITY	LPG	WATER	ALL UTILITIES
TOTAL COST	QAR 4,010,615	QAR 208,503	QAR 1,129,263	QAR 5,348,381
MAXIMUM	QAR 492,639 (Aug)	QAR 23,016 (Feb)	QAR 116,029 (Jun)	QAR 614,287 (Jul)
AVERAGE	QAR 334,218	QAR 17,375	QAR 94,105	QAR 445,698
MINIMUM	QAR 248,298 (Mar)	QAR 12,275 (Nov)	QAR 75,716 (Dec)	QAR 316,523 (Feb)
PERCENTAGE	75%	4%	21%	100%

Figure 4: Annual Utilities Costs Breakdown



3. Utilities Energy Summary

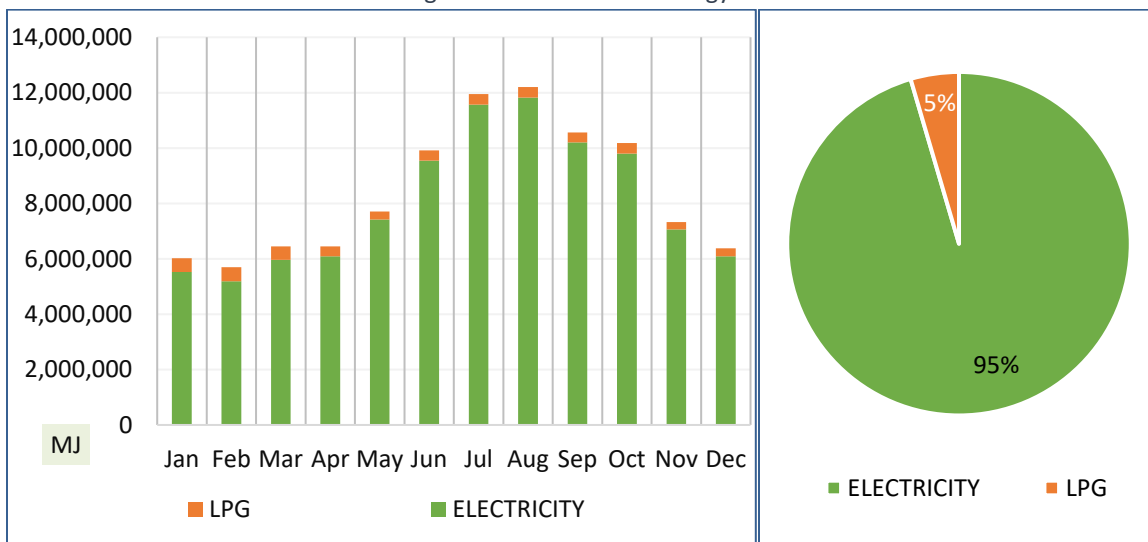
3.1 Utilities Energy Breakdown

To be able to aggregate all the utilities energy consumption, the energy consumption of all energy sources is converted to a single energy unit MJ (mega joules). The breakdown for the energy sources consumption is shown in the following table. As can be seen, electricity accounts for 95% of the energy sources energy consumption, whereas Gas only accounts for 5%.

Table 2: Total Utilities Energy Breakdown

MONTH	ELECTRICITY	LPG	TOTAL
Unit	MJ	MJ	MJ
Jan	5,523,750	498,640	6,022,390
Feb	5,185,112	506,358	5,691,471
Mar	5,959,163	492,042	6,451,205
Apr	6,088,450	363,847	6,452,296
May	7,413,188	293,440	7,706,628
Jun	9,542,783	373,333	9,916,116
Jul	11,567,970	376,955	11,944,925
Aug	11,823,347	383,509	12,206,856
Sep	10,202,846	355,050	10,557,896
Oct	9,796,979	388,511	10,185,490
Nov	7,056,140	270,061	7,326,201
Dec	6,095,023	285,325	6,380,348
Total	96,254,752	4,587,071	100,841,822
Maximum	11,823,347 (Aug)	506,358 (Feb)	12,206,856 (Aug)
Average	8,021,229	382,256	8,403,485
Minimum	5,185,112 (Feb)	270,061 (Nov)	5,691,471 (Feb)
Percentage	95%	5%	100%

Figure 5: Total Utilities Energy Breakdown



As per the above table, the **annual energy footprint** of FSD stands at: **2,088.34 MJ/m²**

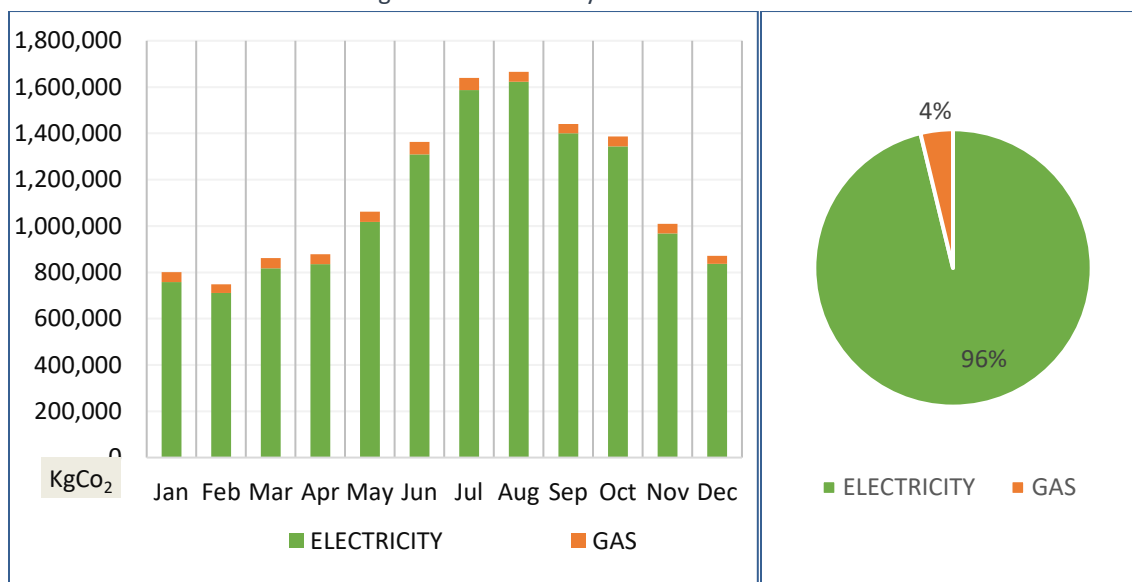
3.2 FSB Carbon Emissions and Footprint

The carbon emissions calculation is done by multiplying the consumption of each utility (electricity and gas) by its emissions factor. The electricity emissions factor varies between countries, with the Qatar average of 0.49 KgCO₂/KWh being taken in this report. LPG emission factor is 2.78 KgCO₂/m³. The **total annual carbon emissions** are **13,726,789 KgCO₂**.

Table 3: FSB Carbon Emissions Monthly Breakdown

MONTH	ELECTRICITY	GAS	TOTAL
Unit	KgCO ₂	KgCO ₂	KgCO ₂
Jan	757,995	42,394	800,389
Feb	711,525	36,328	747,854
Mar	817,744	44,201	861,945
Apr	835,486	42,294	877,780
May	1,017,273	44,811	1,062,084
Jun	1,309,505	53,064	1,362,569
Jul	1,587,411	52,085	1,639,496
Aug	1,622,455	43,986	1,666,441
Sep	1,400,082	40,932	1,441,015
Oct	1,344,387	42,020	1,386,407
Nov	968,277	40,831	1,009,107
Dec	836,388	35,314	871,702
Total	13,208,529	518,260	13,726,789
Maximum	1,622,455 (Aug)	53,064 (Jun)	1,666,441 (Aug)
Average	1,100,711	43,188	1,143,899
Minimum	711,525 (Feb)	35,314 (Dec)	747,854 (Feb)
Percentage	96%	4%	100%

Figure 6: FSB Monthly Carbon Emissions



As per the above table, the **annual carbon intensity** of FSD stands at: **284.27 KgCO₂/m²**.

4. Electrical Energy Balance

The Electrical Energy Balance provides the breakdown per chosen end-use category for both the installed electrical loads (KW) and the annual energy consumed (KWh). This is achieved following the energy simulation and bills reconciliation completion as per the load inventory, the related operations and the inspection done during the field energy audit.

In this section, the results of two different energy balances will be shown:

1. The energy balance based on the electrical consumptions of all the FSD complex components (excluding the two residential towers) with 100% of the chilled water plant operation considered. This balance was used in the load categories description, which is the section above, and will be used in the energy efficiency measures savings calculations.
2. The energy balance based on the electrical consumptions of all the FSD complex components (excluding the two residential towers) with 75% of the chilled water plant operation considered, since the chilled water produced by the plant is shared between the two residential towers and the rest of the complex, with the towers taking 25% of the total chilled water production, and the rest of the complex the remaining 75%.

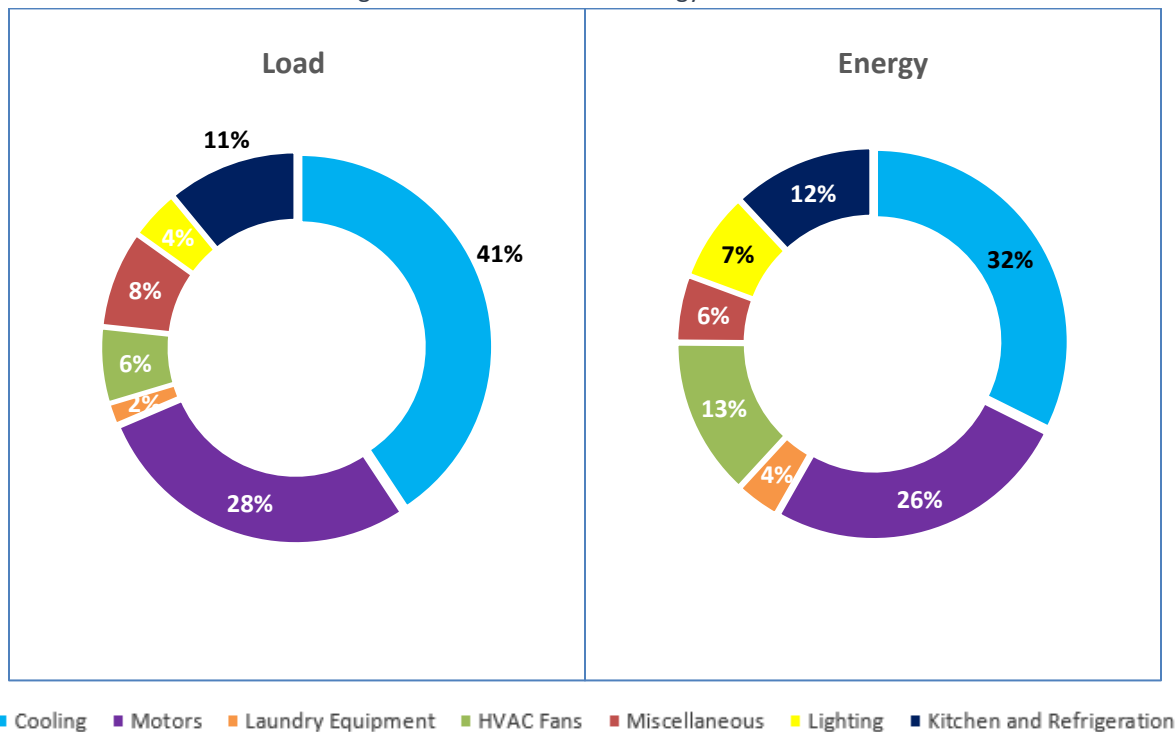
3.1.1. Energy Balance with 100% Chilled Water Plant Operation

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and annual energy consumption in KWh.

Table 4: Electrical Load & Energy Consumption Breakdown by Category

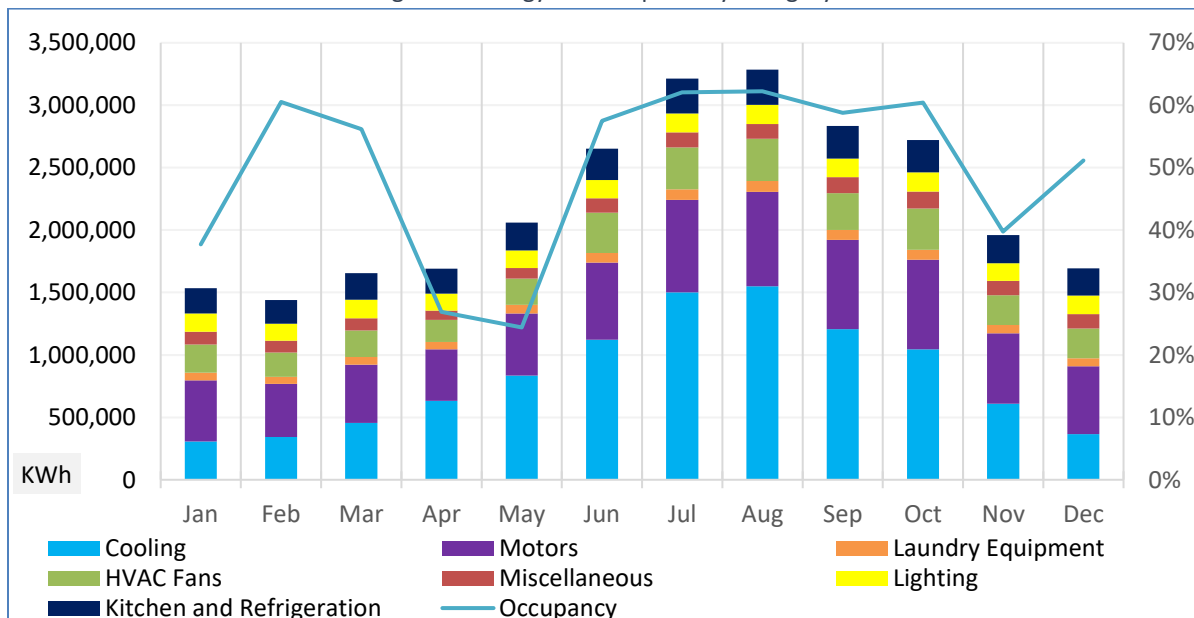
Category	Load		Energy			
Unit	KW	%	KWh	QAR	Kg. CO ₂	%
Cooling	4,009	41%	9,970,102	QAR 4,486,546	6,829,520	37%
Motors	2,745	28%	6,942,608	QAR 3,124,174	4,755,686	26%
Laundry Equipment	177	2%	850,536	QAR 382,741	582,617	3%
HVAC Fans	624	6%	3,120,993	QAR 1,404,447	2,137,881	12%
Miscellaneous	804	8%	1,302,368	QAR 586,066	892,122	5%
Lighting	410	4%	1,748,067	QAR 786,630	1,197,426	7%
Kitchen and Refrigeration	1,080	11%	2,802,757	QAR 1,261,241	1,919,889	10%
Total	9,848	100%	26,737,431	QAR 12,031,844	18,315,140	100%
Maximum	Cooling - 4,009 KW, 41%		Cooling - 9,970,102 KWh, AED 4,486,546, 37%			
Minimum	Laundry Equipment - 177 KW, 2%		Laundry Equipment - 850,536 KWh, AED 382,741, 3%			

Figure 7: Electrical Load & Energy Breakdown



The Cooling category shows the highest installed load at 41% of the total, followed by the Motors at 28%. Whereas the Cooling category also has the highest share of energy consumption at 37% of the total, followed by the Motors category at 26%.

Figure 8: Energy Consumption by Category



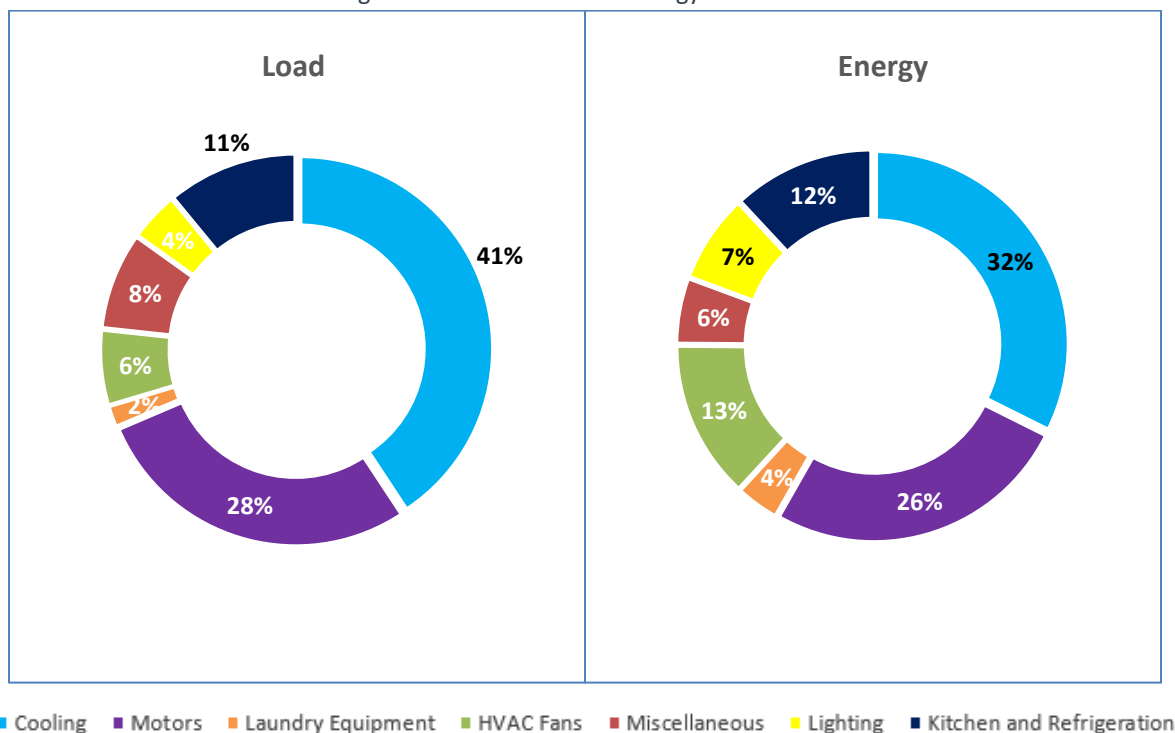
3.1.2. Energy Balance with 75% Chilled Water Plant Operation

The following table and figure provide the percentage shares of categories' total installed electrical power in KW and annual energy consumption in KWh.

Table 5: Electrical Load & Energy Consumption Breakdown by Category

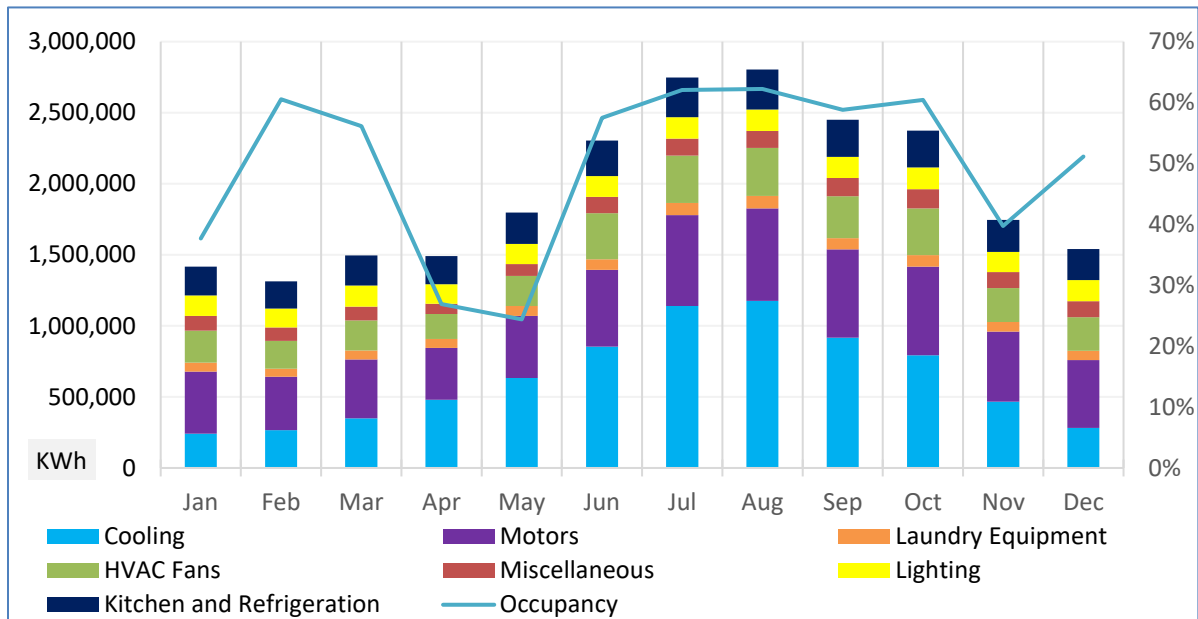
Category	Load		Energy			
Unit	KW	%	KWh	QAR	Kg. CO ₂	%
Cooling	4,009	41%	7,601,394	QAR 3,420,627	5,206,955	32%
Motors	2,745	28%	6,073,646	QAR 2,733,141	4,160,448	26%
Laundry Equipment	177	2%	850,536	QAR 382,741	582,617	4%
HVAC Fans	624	6%	3,117,640	QAR 1,402,938	2,135,583	13%
Miscellaneous	804	8%	1,296,251	QAR 583,313	887,932	6%
Lighting	410	4%	1,744,575	QAR 785,059	1,195,034	7%
Kitchen and Refrigeration	1,080	11%	2,802,757	QAR 1,261,241	1,919,889	12%
Total	9,848	100%	23,486,799	QAR 10,569,060	16,088,457	100%
Maximum	Cooling - 4,009 KW, 41%		Cooling - 7,601,394 KWh, 32%			
Minimum	Laundry Equipment - 177 KW, 2%		Laundry Equipment - 850, 4%			

Figure 9: Electrical Load & Energy Breakdown



The Cooling category shows the highest installed load at 41% of the total, followed by the Motors at 28%. Whereas the Cooling category also has the highest share of energy consumption at 32% of the total, followed by the Motors category at 26%.

Figure 10: Energy Consumption by Category



5. Thermal Energy

FSD's thermal energy consists of hot water production through six LPG driven boilers and steam production through two LPG steam boilers.

The hot water system in FSD is composed of 6 LPG driven boilers, with each two serving a zone. The hotel has 3 different zones: the 3rd to 10 floor and the 11th to 17th floor which are served by four identical boilers (two for each zone), while the third zone which consists of the lower level to the 2nd floor are served by two larger boilers, since it includes the laundry and the main kitchen in the BOH. Each zone has one calorifier and water is maintained at 60°C.

The steam boilers system is exclusively used for the laundry and pool heating through heat exchangers and have respectively a total capacity of 1. 25 tons/hr and 1849KW working at a pressure of 150 Psi

Figure 11: Hot Water Boilers (Left) and Steam Boilers(Right)



There are no submetering of LPG at the level of usage between the Hot Water and Steam Boilers. The hotel tracks only the total Hotel consumption and the NOBU restaurant separately with a monthly share of 5% to 7% across the year.

The 93% to 95% of the remaining consumption is divided between the kitchens in the Hotel building as well as the Steam and Hot Water boilers with the kitchens potentially representing about 25% of the total or an estimated 25,000m³ of LPG annually with a related cost of 50,000 QAR.

The boilers are estimated to consume annually about 75,000m³ with a related cost of 148,000 QAR

6. Energy Efficiency Measures

The below section provides the roadmap that would include a number of Energy Efficiency Measure (EEM) which are all the actions that would reduce the utilities consumption. In this aim, EEG investigated a considerable number of energy efficiency solutions, which could yield to substantial energy cost and carbon footprint reductions in the event of their implementation. These opportunities include those who require very little or no CAPEX at all – Low Cost/No Cost – [LC/NC] as well as those needing capital expenditure [CAPEX]. The assessment of an EEM is its financial feasibility and technical viability. Each EEM is attributed with a color code which is a priority or urgency indicator for the EEM's implementation. The color codes adopted in this report along the proposed measures are found in the below tables:

Color Code	Description	Priority
	Measures with very small CAPEX and excellent payback (<1 year) – Low Cost/No Cost	A
	Measures with good payback period (<3 years)– Immediate Implementation	A
	Measures with midterm payback period (3 to 6 Years) – Strategic Implementation	B
	Measures to be done during a planned retrofit/replacement	C
	Measures with high payback period (>6 years) – Upon CAPEX Availability	D

Table 5: Energy Efficiency Measures

CODE	TYPE	TITLE	
EEM - 01	LCNC	Lighting Management	Control lighting in function areas
EEM - 02	CAPEX	Nobu Restaurant & SPA Lighting Retrofit	Replace existing lighting fixtures by LED equivalent
EEM - 03	CAPEX	Motion Detectors Installation	Install Motion Detectors in technical rooms and individual offices
EEM - 04	LCNC	HVAC Control Optimization	Scheduled operation implementation, AHU temperature reset, proactive VFD operation and Winter mode implementation across all units.
EEM - 05	CAPEX	VFD Installation on Pool Pumps	Install VFDs on pool pumps
EEM - 06	CAPEX	SPA Building Load Management	Lighting & Equipment Management , HVAC Control & covering pools
EEM - 07	LCNC	Chilled Water Plant Operation Optimization	Implement a comprehensive Chillers Plant Control Optimization : Proactive DP setting, CW temp. reset, Condenser Water temp. reset, Chillers sequencing
EEM - 08	CAPEX	VFD Installation on Condenser Pumps	Install VFDs on condenser pumps
EEM - 09	CAPEX	Sea Water Condenser Cooling	Shift condenser cooling from Cooling Tower to Sea Water
EEM - 10	CAPEX	Photovoltaic Driven Electricity Generation	Install 114KW PV Plant for electricity production
EEM - 11	LCNC	Notes on Residential Tower Chilled Water	Install BTU meters for equitable charging and coordinate to proactively control tertiary pumps

The following table provides the summary of the Efficiency Measures with key indicators including CAPEX, % Savings from Total Utilities Cost and Payback period.

As can be seen, if all EEMs are implemented, the total annual cost savings would be **QAR1.27M** at a CAPEX of **QAR 4.9M** and a payback of **3.9 Years** (including the LC/NC EEMs which had savings of **QAR282K** mostly related to the HVAC Control and Chilled Water Plant Optimizations.

Clearly all payback periods will be impacted by any changes in tariffs in the future .

Table 7: Energy Efficiency Measures – Financial Summary

CODE	EEM Name	ANNUAL SAVINGS		COST	PAYBACK
-	-	QAR	KGs CO ₂	QAR	Years
EEM - 01 [LCNC]	Lighting Management	QAR 8,038.33 (0.2%)	26,473	QAR 0.00	-
EEM - 02 [CAPEX]	Nobu Restaurant & SPA Lighting Retrofit	QAR 42,600.00 (0.9%)	140,299	QAR 169,000.00	4.0
EEM - 03 [CAPEX]	Motion Detectors Installation	QAR 27,535.65 (0.6%)	90,686	QAR 18,000.00	0.7
EEM - 04 [LCNC]	HVAC Control Optimization	QAR 124,701.50 (2.6%)	410,691	QAR 0.00	-
EEM - 05 [CAPEX]	VFD Installation on Pool Pumps	QAR 10,827.29 (0.2%)	35,659	QAR 46,600.00	4.3
EEM - 06 [CAPEX]	SPA Building Load Management	QAR 27,000.00 (0.6%)	0	QAR 44,400.00	1.6
EEM - 07 [LCNC]	Chilled Water Plant Operation Optimization	QAR 150,146.43 (3.1%)	494,491	QAR 0.00	-
EEM - 08 [CAPEX]	VFD Installation on Condenser Pumps	QAR 67,292.16 (1.4%)	221,619	QAR 200,000.00	3.0
EEM - 09 [CAPEX]	Sea Water Condenser Cooling	QAR 516,876.75 #REF!	347,206	QAR 2,200,000.00	4.3
EEM - 10 [CAPEX]	Photovoltaic Driven Electricity Generation	QAR 294,940.20 (6.1%)	971,354	QAR 2,282,280.00	7.7
	TOTAL	QAR 1,269,958.30 (26.1%)	2,738,478	QAR 4,960,280.00	3.9

(EEM#11 was added in the report as a note regarding the residential towers chilled water line with two recommendations)

The below table provides the annual saving for electricity and water. Electricity savings are seen to be **5.7MWh** per year representing **24%** of the actual electricity consumption (including EEM 10/PV Power Plant), while water Consumption will be reduced by **33%** in volume due to the sea water cooling project (EEM-09) that would cancel all the TSE water used for the cooling towers.

Note that various EEMs were considered for the gas end uses (hot water and steam boilers) and were not found economically attractive due to the low gas prices in Qatar. Potential measures included the installation of solar thermal for hot water production and heat recoveries from the steam boilers and laundry's exhausts. Payback period were more than 7 years,

Table 8: Energy Efficiency Measures – Consumption Savings Summary

CODE	EEM Name	ELECTRICITY		Water	
-	-	KWh		m ³	
EEM - 01	Lighting Management	53,589	(0.2%)	0	(0.0%)
EEM - 02	Nobu Restaurant & SPA Lighting Retrofit	284,000	(1.2%)	0	(0.0%)
EEM - 03	Motion Detectors Installation	183,571	(0.8%)	0	(0.0%)
EEM - 04	HVAC Control Optimization	831,343	(3.5%)	0	(0.0%)
EEM - 05	VFD Installation on Pool Pumps	72,182	(0.3%)	0	(0.0%)
EEM - 06	SPA Building Load Management	180,000	(0.8%)	0	(0.0%)
EEM - 07	Chilled Water Plant Operation Optimization	1,000,976	(4.3%)	0	(0.0%)
EEM - 08	VFD Installation on Condenser Pumps	448,614	(1.9%)	0	(0.0%)
EEM - 09	Sea Water Condenser Cooling	702,833	(3.0%)	125,839	(33.1%)
EEM - 10	Photovoltaic Driven Electricity Generation	1,966,268	(8.4%)	0	(0.0%)
TOTAL		5,723,377	24%	125,839	33%

END of EXECUTIVE SUMMARY